Project Initialization and Planning Phase

Date	10 July 2024
Team ID	xxxxxx
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to develop an intelligent flood prediction system using Machine Learning to improve disaster preparedness and reduce the impact of floods. By analyzing historical rainfall and meteorological data, the system predicts the likelihood of flood events with high accuracy. The solution integrates the trained machine learning model into a Flask-based web application, enabling disaster management authorities, meteorologists, and residents to receive timely flood risk predictions and support informed decision-making.

Project Overview	
Objective	Develop an intelligent Machine Learning-based Flood Prediction System capable of accurately predicting flood risk using historical rainfall and weather data. The system aims to support early warning, improve disaster preparedness, and assist authorities in making timely decisions during extreme weather conditions.
Scope	The project includes collecting historical rainfall and weather datasets, performing data preprocessing and exploratory data analysis, training multiple machine learning classification models, selecting the best-performing model, and deploying it through a Flask web application for real-time flood prediction.
Problem Statement	
Description	Floods are among the most destructive natural disasters, causing significant loss of life, property damage, and environmental disruption. Existing flood forecasting methods often lack sufficient accuracy and timely prediction, making disaster preparedness challenging. There is a need for an intelligent system that can accurately predict flood risk using historical weather and rainfall data.
Impact	The proposed solution enables early flood warnings, improves disaster preparedness, supports efficient resource allocation, and helps reduce the social and economic impact of floods by providing accurate and timely predictions.
Proposed Solution	

Approach	Develop a supervised Machine Learning model using historical rainfall and meteorological data. Train and compare multiple classification algorithms, including **Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost**, and integrate the best-performing model into a Flask web application for real-time flood prediction.
Key Features	☐ Machine Learning-based flood risk prediction. ☐ Historical rainfall and weather data analysis. ☐ Data preprocessing and feature engineering. ☐ Comparison of multiple classification algorithms. ☐ High-accuracy flood prediction model. ☐ Interactive Flask web application. ☐ User-friendly prediction interface. ☐ Scalable deployment for disaster management applications.

	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv